Lecture 11

1. Vector Space:

Def: A vector space is a non-empty set $\bar{V}$ whose objects we will call vectors, with a set of scalars and a way to add vectors

The addition satisfies the following:

If $\vec{u}, \vec{w}, \vec{v} \in \bar{V}$,

A1: $\vec{u} + \vec{v} \in \bar{V}$

A2: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

A3: $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$

A4: There is a zero vector $\vec{0}$ such that

$$\vec{0} + \vec{v} = \vec{v} \text{ for all } \vec{v} \in \bar{V}$$

A5: For each $\vec{v} \in \bar{V}$ there is a vector $-\vec{v}$ such that $\vec{v} + (-\vec{v}) = \vec{0}$

The multiplication satisfies the following:

If $c, d \in \mathbb{R}$

M1: $c\vec{v} \in \bar{V}$

M2: $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$

M3: $(c + d)\vec{v} = c\vec{v} + d\vec{v}$

M4: $c(d\vec{v}) = (cd)\vec{v}$

M5: $1\vec{v} = \vec{v}$

The: Let $\bar{V}$ be a vector space. For any $\vec{v} \in \bar{V}$ and scalar c we have

$$0\vec{v} = \vec{0}$$

$$c\vec{0} = \vec{0}$$

$$-\vec{v} = (-1)\vec{v}$$

2. Subspaces:

Def: A subspace of a vector $\bar{V}$ is a subset H of $\bar{V}$ such that

1. $\vec{0} \in H$
2. If $\vec{u}, \vec{v} \in H$ then $\vec{u} + \vec{v} \in H$
3. If c is a scalar and $\vec{v} \in H$ then $c\vec{v} \in H$

The: Any vector space $\bar{V}$ has the following two subspaces

- The zero subspace which only has $\vec{0}$ in it $\{\vec{0}\}$
- $\bar{V}$ is the subspace of itself

3. Subspaces Spanned by a set

Def: The subspace generated by $\vec{v_1}, \vec{v_2}, \dots, \vec{v_n}$ or the subspace spanned by $\vec{v_1}, \vec{v_2}, \dots, \vec{v_n}$ is $\text{Span}(\vec{v_1}, \dots, \vec{v_n})$. If H is a subspace of V and we find vectors $\vec{v_1}, \dots, \vec{v_n}$ such that $H = \text{Span}(\vec{v_1}, \dots, \vec{v_n})$ then we say that the vectors $\vec{v_1}, \dots, \vec{v_n}$ are a generating set or a spinning set for H and we say $\{\vec{v_1}, \dots, \vec{v_n}\}$ generates or spans H